Lab Standard: SLURM Scripts for Scalable ML Experiments

Medical Computer Vision Lab (MCV Lab)

October 16, 2026

1 Purpose

This document defines the lab standard for running ML experiments on HPC clusters using SLURM. All jobs must be:

- reproducible
- logged
- checkpointed
- sweep-ready

2 General Rules

S1. Never edit training scripts in HPC jobs. Use configs/overrides only.

S2. Always log SLURM environment variables.

S3. Always write outputs into a run folder.

S4. Always checkpoint (last + best).

S5. Always specify GPU, CPU, RAM explicitly.

3 Folder Standard

SLURM logs must be stored separately from run outputs:

```
project/
  runs/
  slurm_logs/
    %x_%j.out
    %x_%j.err
  scripts/
    run_train.slurm
```

4 Baseline SLURM Job (Single Run)

Save as `scripts/run_train.slurm`:

```
#!/bin/bash
#SBATCH --job-name=mcv_train
#SBATCH --output=slurm_logs/%x_%j.out
#SBATCH --error=slurm_logs/%x_%j.err
#SBATCH --time=08:00:00
#SBATCH --partition=gpu
```

```
#SBATCH --gres=gpu:1
#SBATCH --cpus-per-task=8
#SBATCH --mem=32G

# Optional: email notifications
# SBATCH --mail-type=END,FAIL
# SBATCH --mail-user=your_email@uvic.ca

set -euo pipefail

echo "=== SLURM JOB INFO ==="
echo "JOB_ID:      ${SLURM_JOB_ID}"
echo "JOB_NAME:    ${SLURM_JOB_NAME}"
echo "NODELIST:    ${SLURM_NODELIST}"
echo "GPUS:        ${SLURM_GPUS:-unknown}"
echo "CPUS:        ${SLURM_CPUS_PER_TASK}"
echo "WORKDIR:     ${SLURM_SUBMIT_DIR}"
echo "START_TIME:  $(date)"

cd ${SLURM_SUBMIT_DIR}

# Activate environment
# Option A: conda
# source ~/miniconda3/etc/profile.d/conda.sh
# conda activate mcv

# Option B: module-based
# module load python/3.10 cuda/12.1
# pip install -r requirements.txt

nvidia-smi || true

# Run training
python train.py \
  --dataset rsna_trauma \
  --model resnet50 \
  --lr 1e-4 \
  --seed 0 \
  --amp \
  --out_dir runs \
  --run_name "rsna_resnet50_seed0_lr1e4_${SLURM_JOB_ID}"

echo "END_TIME: $(date)"
```

5 SLURM Array for Sweeps

SLURM arrays are the lab standard for running sweeps.

5.1 Example: Sweep over seeds

Save as `scripts/sweep_seeds.slurm`:

```
#!/bin/bash
#SBATCH --job-name=mcv_seed_sweep
#SBATCH --output=slurm_logs/%x_%A_%a.out
#SBATCH --error=slurm_logs/%x_%A_%a.err
#SBATCH --time=08:00:00
```

```
#SBATCH --partition=gpu
#SBATCH --gres=gpu:1
#SBATCH --cpus-per-task=8
#SBATCH --mem=32G
#SBATCH --array=0-4

set -euo pipefail
cd ${SLURM_SUBMIT_DIR}

SEEDS=(0 1 2 3 4)
SEED=${SEEDS[$SLURM_ARRAY_TASK_ID]}

python train.py \
  --dataset rsna_trauma \
  --model resnet50 \
  --lr 1e-4 \
  --seed ${SEED} \
  --amp \
  --out_dir runs \
  --run_name "rsna_resnet50_seed${SEED}_job${SLURM_JOB_ID}"
```

Submit:

```
sbatch scripts/sweep_seeds.slurm
```

6 Hydra + SLURM Recommended Pattern

If using Hydra multirun, we prefer either:

- Hydra `-m` multirun inside a single SLURM job (small sweeps)
- SLURM array + Hydra overrides (large sweeps; recommended)

6.1 SLURM array + Hydra override template

```
python train.py \
  dataset=rsna_trauma \
  model=resnet50 \
  optim=adamw \
  optim.lr=1e-4 \
  seed=${SEED}
```

7 Multi-GPU Training (DDP)

7.1 Standard: torchrun

Save as `scripts/ddp_train.slurm`:

```
#!/bin/bash
#SBATCH --job-name=mcv_ddp
#SBATCH --output=slurm_logs/%x_%j.out
#SBATCH --error=slurm_logs/%x_%j.err
#SBATCH --time=12:00:00
#SBATCH --partition=gpu
#SBATCH --gres=gpu:2
#SBATCH --cpus-per-task=12
#SBATCH --mem=64G
```

```
set -euo pipefail
cd ${SLURM_SUBMIT_DIR}

export OMP_NUM_THREADS=${SLURM_CPUS_PER_TASK}

# Use torchrun for DDP
torchrun --nproc_per_node=2 train.py \
  --dataset rsna_trauma \
  --model resnet50 \
  --lr 1e-4 \
  --seed 0 \
  --amp \
  --out_dir runs \
  --run_name "rsna_ddp2_${SLURM_JOB_ID}"
```

8 Checklist for SLURM Job Reviews

Before using a SLURM script in the lab:

- Logs go to `slurm_logs/`
- Run outputs go to `runs/`
- Script logs SLURM metadata
- Script runs without editing python source
- Seeds are parameterized
- Checkpoints saved (best + last)